



December 2025

EC3002-1 – COMPUTER NETWORKS AND CYBER SECURITY
(For EC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. A layer at one level on the sender side of a network communicates with layers at the _____ level on the receiver side.
A) higher
B) lower
C) corresponding
D) Medium
2. Which category of HDLC frames undergoes error and flow control mechanisms by comprising send and receive sequence numbers?
A) U-frames
B) I-frames
C) S-frames
D) M-frames
3. The maximum throughput for pure ALOHA is _____ per cent.
A) 12.2
B) 18.4
C) 1.84
D) 1.22
4. The vulnerable time for CSMA is _____ the propagation time.
A) same as
B) two times
C) three times
D) half of
5. Which of the following are transport layer protocols.
A) TCP and UDP
B) IP
C) HTTPS
D) HTTP
6. Which ARQ mechanism deals with the transmission of only damaged or lost frames despite the other multiple frames by increasing the efficiency & its utility in noisy channels?
A) Go-Back-N ARQ
B) Selective Repeat ARQ
C) Stop-and-Wait ARQ
D) Stop-and-go ARQ
7. The session layer is responsible for _____ and _____.
A) segmentation and reassembly.
B) dialog control and synchronization
C) network and terminal
D) error control and flow control
8. _____ protocol is used to associate a logical address with a physical address.
A) Reverse Address Resolution
B) Internet Control Message
C) Internet Group Message
D) Address Resolution
9. A _____ can check the MAC addresses contained in the frame.
A) passive hub
B) repeater
C) bridge
D) router
10. A _____ normally connects LANs and WANs in the Internet and has a table that is used for making decisions about the route.
A) repeater
B) bridge
C) router
D) hub
11. In IPv4 header, an HLEN value of decimal 10 means _____.
A) there are 10 bytes of options
B) there are 40 bytes of options
C) there are 10 bytes in the header
D) there are 40 bytes in the header

12. The class of the IPv4 address: 4.5.6.7 is,
 A) A B) B
 C) C D) D
13. If one of the addresses in a block of classless address is 12.2.2.76/27, then the first address of that block is,
 A) 12.2.2.0 B) 12.2.2.32
 C) 12.2.2.64 D) 12.2.2.8
14. An organization is granted a block; one address is 2.2.2.64/20. The organization needs 10 subnets. Then the subnet prefix length is,
 A) /20 B) /24
 C) /25 D) /27
15. _____ refers to the way a packet is delivered from one station to the next station.
 A) Forwarding B) Delivery
 C) Routing D) Segmentation
16. The value of acknowledgement field in a segment defines _____.
 A) sequence number of the byte received previously B) total number of bytes to receive
 C) sequence number of the next byte to be received D) sequence of zeros and ones
17. The algorithm primarily used for digital signatures is,
 A) AES B) RSA
 C) DES D) Blowfish
18. Confidentiality of a message is ensured by,
 A) Encryption B) Decryption
 C) Hashing D) Digital signature
19. Attack through which an attacker inserts malicious code into a website to steal information from users is,
 A) SQL Injection B) Cross-site scripting
 C) Dos D) Phishing
20. In SSL, the protocol used to establish the initial connection and negotiate encryption parameters is,
 a) protocol
 b) Change Cipher Spec protocol
 c) Alert protocol
 d) Record protocol
 A) Handshake B) Change Cipher
 C) Alert D) Record

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

	Marks	BT*	CO*	PO*
1. a) Name and explain the four levels of addresses used in the internet model.	6	L*2	1	1
b) Communication devices in a network are needed to connect using (i) Mesh, (ii) Star, (iii) Bus. For each of these show the connections made and explain the consequences if a connection fails.	6	L2	1	1
c) Given the data word 10011 and the predefined CRC generator polynomial 10001000000100001. What is the actual message string that is transmitted by the sender node using CRC coding technique?	4	L3	2	1
2. a) Explain the responsibilities of the physical layer, data link layer and network layer in the Internet model?	6	L2	1	1
b) With the help of a flow diagrams explain why send window size in Go-back-N automatic repeat request protocol should be less than 2^m . Where m is the number of bits in sequence number.	6	L3	2	1

- c) The following frame is received by a destination machine which uses bit-oriented protocol. Write the exact data that has been forwarded to the network layer.

Flag	Header	00011111011001111100100011110	Trailer	Flag
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4 L3 2 1

3. a) Compare ISO-OSI and TCP/IP internet models.
 b) Write the name of the noisy channel protocol in which several frames are sent before receiving an acknowledgement from the receiver. Explain receiver site algorithm employed in that protocol.
 c) With the help of a neat diagram explain ring topology of a given computer network. Mention advantages and disadvantages of ring topology.

6 L2 1 1

6 L2 2 1

4 L2 1 1

Unit – II

4. a) What characteristic can be used to group stations in a VLAN? Explain.
 b) Explain how bandwidth of a link is shared between different stations using frequency division and time division multiple access methods. Support your answer with suitable diagrams.
 c) A slotted ALOHA network transmits 200-bit frames using a shared channel with a 200-kbps bandwidth. Find the throughput and number of frames those get survived, if the system (all stations together) produces, (i) 500 frames per second, and (ii) 250 frames per second.
5. a) Name the three persistence methods used by communication channel to avoid data collision and explain them using flow diagrams.
 b) Name and explain the five different categories of the connecting devices based on the layers in which they operate in the network.
 c) How many numbers of bits are used to denote an IPv4 address? Find the class of the following IP addresses.
 (i) 208.34.54.12
 (ii) 238.34.2.1
 (iii) 114.34.2.8
6. a) What is a transparent bridge? Outline the three criteria that transparent bridges must meet.
 b) For the network shown in the fig. Q6(b), let A is the root node. Find the shortest path using Dijkstra algorithm.

6 L2 4 1

6 L2 3 1

4 L3 3 1

6 L2 3 1

6 L2 4 1

4 L3 4 1

6 L2 4 1

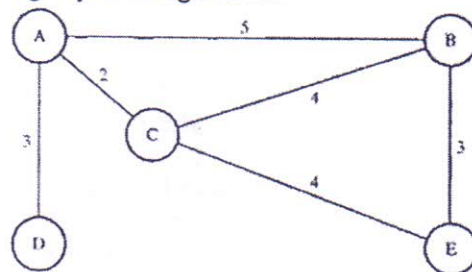


Fig. Q6(b)

6 L3 4 1

- c) The following is a dump of a UDP header in hexadecimal format:
 0632000D001CE217
 (i) What is the source port number?
 (ii) What is the destination port number?
 (iii) What is the total length of the user datagram?
 (iv) What is the length of the data?

4 L3 4 1

Unit – III

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|-------|---|---|----|---|---|
| 7. a) | What is cyber stalking? How cyber stalking works? | 6 | L2 | 5 | 1 |
| b) | List and explain any six ID theft counter measures. | 6 | L2 | 5 | 1 |
| c) | Explain symmetric and asymmetric cryptography methods. | 4 | L2 | 5 | 1 |
| 8. a) | What is social engineering? Briefly explain with an example, the two types of social engineering. | 6 | L2 | 5 | 1 |
| b) | What is a firewall? Explain its functions. | 6 | L2 | 5 | 1 |
| c) | Explain the code conversion and segmentation services provided by Pretty Good Privacy (PGP). | 4 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
